

5	(a)	(i)	Resistance of thermistor is given by the ratio of p.d. across thermistor, V to current through thermistor, I. In Fig. 5.2, the ratio of V/I decreases as current or voltage increases. As current or voltage increases, the thermistor will heat up. Thus, the thermistor's resistance decreases with increasing temperature indicating that it is a ntc thermistor. <i>Alternatively,</i> Resistance of thermistor is given by the ratio of p.d. across thermistor, V to current through thermistor, I. In Fig. 5.2, the gradient of the straight line that joins the origin to a point on the I-V graph of thermistor increases, the reciprocal of this gradient gives the ratio of V/I, i.e. V/I decreases as current or voltage increases. As current or voltage increases, the thermistor will heat up. Thus, the thermistor's resistance decreases with increasing temperature indicating that it is a ntc thermistor.	B1 B1 Or B1 B1
		(ii)	From the graph, when current through battery is 8.0 A, 1. the current through the filament bulb = 5.5 A 2. the current through the thermistor = 2.5 A 3. the e.m.f. of the battery = 8.0 V	A1 A1 A1
		(b) (i)	Correct sketch of a straight line passing through the origin with gradient = $1/2.5 = 0.4$. Straight line pass through (2.5, 1.0) and (10, 4).	A1
		(ii)	the current through the filament bulb = 4.0 A	A1
		(iii)	the potential difference across the resistor = $4.0 \times 2.5 = 10$ V	A1
			Total marks	8

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6	(a)	(i)	$\sqrt{N}$	(2500)	
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